Rishabh Agarwal

Embedded Firmware Engineer

RESUME_1954

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Total Experience: 12 years

PROFESSIONAL SUMMARY

Embedded Firmware Engineer with 12 years of experience in embedded firmware, using C, C++, ARM Cortex-M, FreeRTOS. Experienced in real-time systems, hardware debugging and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: FreeRTOS, STM32, Git, SPI, Firmware Debugging, UART, C++, C, ARM Cortex-M, Embedded Linux, I2C, CAN

Methods: Real-time systems, Hardware debugging, Protocol validation

PROFESSIONAL EXPERIENCE

Lead Embedded Firmware Engineer | OrionPath Technologies | Jan 2022 - Present

- Lowered field-reset incidents by 39% with watchdog recovery and diagnostics
- Improved battery runtime by 18% with peripheral and sleep-state optimization
- Reduced embedded task jitter by 27% using interrupt priority and queue tuning
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Embedded Firmware Engineer | Ironleaf Engineering | Jan 2018 - Dec 2021

- Reduced embedded task jitter by 20% using interrupt priority and queue tuning
- Validated 483 hardware test vectors across protocol and timing corner cases

Associate Embedded Firmware Engineer | QuantaSpring Systems | Jul 2014 - Dec 2017

- Lowered field-reset incidents by 32% with watchdog recovery and diagnostics
- Improved battery runtime by 33% with peripheral and sleep-state optimization

PROJECTS

Low-Power Metering Device | SPI, Firmware Debugging, C++, STM32

- Optimized interrupt-driven acquisition and sleep cycles.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

Bootloader Update Utility | C, FreeRTOS, UART, ARM Cortex-M

- Developed verified firmware transfer and rollback checks.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

EDUCATION

B.Tech in Electronics and Communication Engineering, Bayview Institute of Engineering, 2014
Relevant Methods: Real-time systems, Hardware debugging, Protocol validation

CERTIFICATIONS

ARM Cortex-M Workshop (2026)